10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 1.1 — Complex Numbers

Unit 1 | Solved Exercise | New Book (PECTAA)

Exercise 1.1 (Solved)

1. Simplify the following:

(i) i^5 (ii) i^{16} (iii) $(-i)^{-19}$

(iv) $27i^{-26}$ (v) $i^{11} + i^5$ (vi) $(i^4+i^3+i^2+i)^2$

(vii) $(i^8/i^5)^{-5}$ (viii) $i^{13} \times i^{29}$

Useful values: $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. Every power of i is reduced with the help of $i^4 = 1$.

(i) i^5

$i^5 = i^4 \times i$

$= (i^2)^2 \times i$

$= (-1)^2 \times i$

$= 1 \times i$

$= i$

Answer: i

(ii) i^{16}

$i^{16} = (i^4)^4$

$= (1)^4$

$= 1$

Answer: 1

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(iii) $(-i)^{-19}$

$(-i)^{-19} = \frac{1}{(-i)^{19}}$

$(-i)^{19} = (-1)^{19} \times i^{19}$

$= -(i^{16} \times i^{3})$

$= -(1 \times (-i))$

$= i$

$\therefore (-i)^{-19} = \frac{1}{(-i)^{19}}$

$= \frac{1}{i}$

$= \frac{1}{i} \times \frac{i}{i}$

$= \frac{i}{i^2}$

$= \frac{i}{-1}$

$= -i$

Answer: $-i$

(iv) $27i^{-26}$

$27i^{-26} = \frac{27}{i^{26}}$

$= \frac{27}{(i^4)^6 \times i^2}$

$= \frac{27}{1 \times (-1)}$

$= \frac{27}{-1}$

Answer: -27

(v) $i^{11} + i^5$

$i^{11} = i^8 \times i^3$

$= (i^4)^2 \times (-i)$

$= -i$

$i^5 = i^4 \times i$

$= i$

$\therefore i^{11} + i^5 = (-i) + i$

$= 0$

Answer: 0

(vi) $(i^4 + i^3 + i^2 + i)^2$

$(i^4 + i^3 + i^2 + i)^2 = (1 - i - 1 + i)^2$

$= (1 - 1 - i + i)^2 = (0)^2$

$= 0$

Answer: 0

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(vii) $(i^8/i^5)^{-5}$

$\frac{i^8}{i^5} = i^{8-5}$

$= i^3$

$= -i$

$\therefore \left(\frac{i^8}{i^5}\right)^{-5} = (-i)^{-5}$

$= \frac{1}{(-i)^5}$

$(-i)^5 = (-1)^5 \times i^5$

$= -1 \times i$

$= -i$

$\therefore \frac{1}{-i} = \frac{1}{-i} \times \frac{i}{i}$

$= \frac{i}{-i^2}$

$= \frac{i}{1}$

$= i$

Answer: i

(viii) $i^{13} \times i^{29}$

$i^{13} \times i^{29} = i^{13+29}$

$= i^{42}$

$= (i^4)^{10} \times i^2$

$= (1)^{10} \times (-1)$

$= -1$

Answer: −1

2. Write in terms of i:

(i) $2 + \sqrt{(-4)}$ (ii) $3 - \sqrt{(-7)}$ (iii) $\frac{2}{5} + \frac{\sqrt{(-16)}}{5}$

(iv) $\sqrt{2} - \sqrt{(-3)}$

(i) $2 + \sqrt{(-4)}$

$\sqrt{(-4)} = \sqrt{(4 \times (-1))}$

$= \sqrt{4} \times \sqrt{(-1)}$

$= 2i$

$\therefore 2 + \sqrt{(-4)} = 2 + 2i$

Answer: 2 + 2i

(ii) $3 - \sqrt{(-7)}$

$\sqrt{(-7)} = \sqrt{7} \times \sqrt{(-1)}$

$= \sqrt{7}\, i$

$\therefore 3 - \sqrt{(-7)} = 3 - \sqrt{7}\, i$

Answer: $3 - \sqrt{7}\, i$

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(iii) $\frac{2}{5} + \frac{\sqrt{(-16)}}{5}$

$\sqrt{(-16)} = \sqrt{16} \times \sqrt{(-1)}$

$= 4i$

$\therefore \quad \frac{2}{5} + \frac{\sqrt{(-16)}}{5} = \frac{2}{5} + \frac{4}{5}\,i$

Answer: 2/5 + 4/5 i

(iv) $\sqrt{2} - \sqrt{(-3)}$

$\sqrt{(-3)} = \sqrt{3} \times \sqrt{(-1)}$

$= \sqrt{3}\,i$

$\therefore \quad \sqrt{2} - \sqrt{(-3)} = \sqrt{2} - \sqrt{3}\,i$

Answer: $\sqrt{2} - \sqrt{3}\,i$

3. Find the values of x and y:

(i) $(2x+5) + (y-3)i = 1 + 2i$

(ii) $(3x+2) - (4-y)i = 5 + 3i$

(iii) $(2+i)x + (1-2i)y = 3 + 4i$

(iv) $(1-i)x + (2+i)y = 4 - i$

(v) $(3x-1) + (2y-3)i = 8 + 7i$

Two complex numbers are equal only when their real parts are equal and their imaginary parts are equal.

(i) $(2x + 5) + (y - 3)i = 1 + 2i$

Comparing real parts: $2x + 5 = 1 \implies 2x = -4 \implies x = -2$

Comparing imaginary parts: $y - 3 = 2 \implies y = 5$

Answer: x = −2, y = 5

(ii) $(3x + 2) - (4 - y)i = 5 + 3i$

Real parts: $3x + 2 = 5$

$3x = 3$

$x = 1$

Imaginary parts: $-(4 - y) = 3$

$y - 4 = 3$

$y = 7$

Answer: x = 1, y = 7

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(iii) $(2 + i)x + (1 - 2i)y = 3 + 4i$

$2x + ix + y - 2iy = 3 + 4i$

$(2x + y) + (x - 2y)i = 3 + 4i$

Real parts: $2x + y = 3$ ... (1)

Imaginary parts: $x - 2y = 4$ ... (2)

From (2): $x = 4 + 2y$

Put in (1): $2(4 + 2y) + y = 3$

$8 + 4y + y = 3$

$5y = -5$

$y = -1$

$\therefore\ x = 4 + 2(-1)$

$= 2$

Answer: $x = 2, y = -1$

(iv) $(1 - i)x + (2 + i)y = 4 - i$

$x - ix + 2y + iy = 4 - i$

$(x + 2y) + (-x + y)i = 4 - i$

Real parts: $x + 2y = 4$ ... (1)

Imaginary parts: $-x + y = -1$ ... (2)

Adding (1) and (2): $3y = 3 \implies y = 1$

From (1): $x + 2(1) = 4$

$x = 2$

Answer: $x = 2, y = 1$

(v) $(3x - 1) + (2y - 3)i = 8 + 7i$

Real parts: $3x - 1 = 8$

$3x = 9$

$x = 3$

Imaginary parts: $2y - 3 = 7$

$2y = 10$

$y = 5$

Answer: $x = 3, y = 5$

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